

BENCHMARK: NON-DETECTION

Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460189.9032 +/- 0.0011 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.001 +/- 0.049                 |
| Transit depth (Rp/Rs)^2                           | 9.6e-05 +/- 0.0095 [%]          |
| Orbital Inclination (inc)                         | 87.67 +/- 1.47                  |
| Airmass coefficient 1 (a1)                        | 12.35 +/- 7.77                  |
| Airmass coefficient 2 (a2)                        | -1.72 +/- 0.96                  |
| Scatter in the residuals of the lightcurve fit is | 62.13 %                         |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 1.22                            |
| Optimal Annulus                                   | 4.5                             |
| Transit Duration (day)                            | 0.0783 +/- 0.0028               |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R fit vs published     | 0.001 ± 0.049 vs 0.1301 (deviation 1.38σ) |
| Duration fit vs published | 0.0783 d vs 0.0908 d (deviation 2.34σ)    |
| Mid-time O-C              | 1.1 min                                   |
| Depth SNR                 | 0.0                                       |
| Residual scatter          | 62.13 %                                   |

Light curve

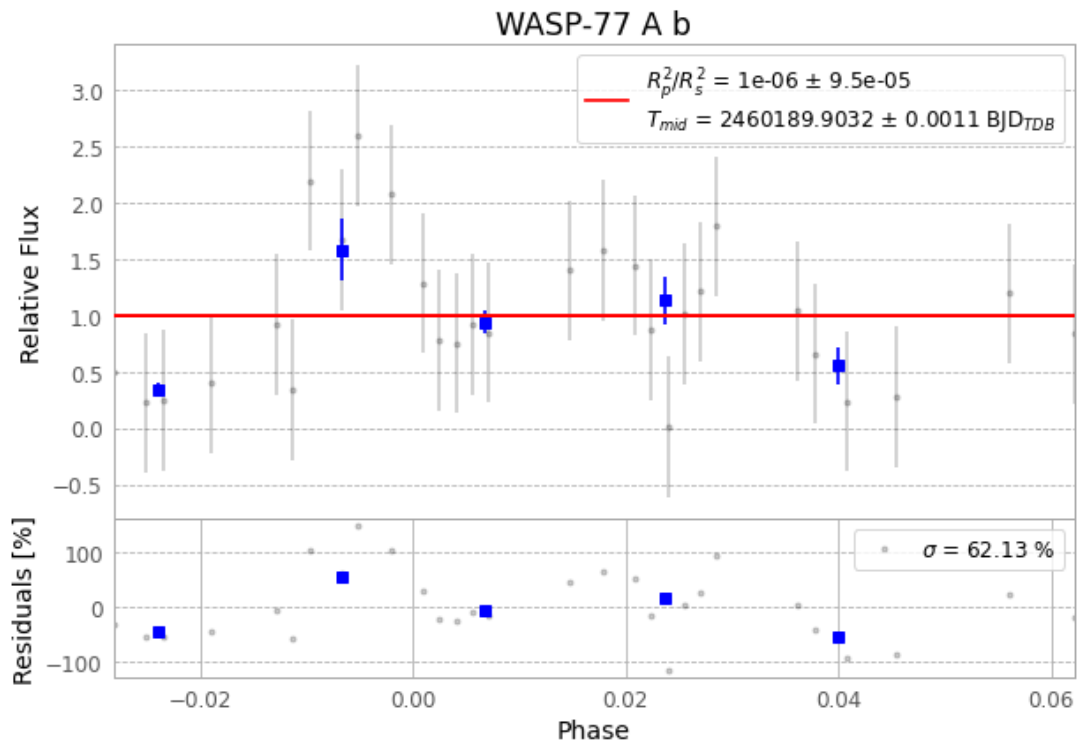


Figure 1 — FinalLightCurve\_WASP-77 A b\_2023-09-02.png (SHA-256 f742caa626754988...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [311, 237], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 18006b00956df792...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

62 FITS frames (41 MB), WASP-77230902083317.FITS → WASP-77230902113915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-77-230902.log (2631b3b5a1a6...), FinalLightCurve\_WASP-77 A b\_2023-09-02.png (f742caa62675...), FinalLightCurve\_WASP-77 A b\_2023-09-02.csv (79d3b289dc7f...)

## Compute resources

Wall time 325.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 17:08Z.